

GULF SHORES, AL, USA

WMO: 720381

Lat: 30.291N

Lon: 87.672W

Elev: 5

StdP: 101.27

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63885

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	0.5	-11.0	1.5	3.2	-7.5	2.0	5.1	10.0	15.3	9.0	16.0	1.7	0	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	34.2	25.9	33.0	25.5	32.3	25.4	28.0	31.7	27.4	30.8	26.8	30.2	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	29.9	26.4	21.9	29.2	26.0	21.3	28.8	89.4	32.0	86.3	30.7	83.9	30.3	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.5	7.6	6.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-4.1	35.6	1.5	2.1	-5.2	37.1	-6.1	38.3	-7.0	39.5	-8.1	41.0		
			-5.2	28.3	1.5	1.1	-6.3	29.1	-7.2	29.7	-8.0	30.4	-9.1	31.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	11.2	12.8	16.0	19.1	23.2	27.0	27.7	27.7	25.9	20.8	15.1	13.3
	DBStd	7.04	5.29	4.97	4.30	3.29	2.80	2.01	1.63	1.72	2.16	4.11	4.44	5.08
	HDD10.0	117	49	27	6	1	0	0	0	0	0	0	8	26
	HDD18.3	825	227	163	96	29	3	0	0	0	0	23	115	168
	CDD10.0	3776	86	105	192	273	408	511	548	549	478	336	162	128
	CDD18.3	1442	6	8	23	52	153	261	290	291	228	101	19	12
	CDH23.3	13919	1	9	48	235	1267	2900	3237	3188	2105	844	75	10
	CDH26.7	5162	0	0	1	8	292	1213	1367	1335	747	196	3	0
Wind	WSAvg	2.7	3.0	3.1	3.4	3.3	2.9	2.6	2.2	2.1	2.2	2.6	2.5	2.9
Precipitation	PrecAvg	1581	129	118	138	119	106	147	187	161	150	92	111	123
	PrecMax	2197	402	249	295	433	338	354	398	253	580	319	297	326
	PrecMin	1050	30	28	14	20	17	34	58	61	40	1	30	49
	PrecStd	294	82	53	77	87	70	82	73	61	106	79	75	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	24.9	26.3	27.7	32.7	36.2	35.4	35.2	33.6	31.9	26.8	24.5
		MCWB	20.4	21.2	20.1	20.6	23.6	24.9	25.8	26.8	25.0	24.8	21.7	21.7
	2%	DB	22.3	22.8	24.9	26.5	30.8	34.3	33.7	33.6	32.3	30.2	25.2	23.3
		MCWB	20.0	19.5	20.1	21.1	23.8	25.3	26.2	26.3	24.8	23.9	20.9	21.1
	5%	DB	21.1	21.4	23.2	25.7	29.4	32.7	32.6	32.6	31.4	28.9	24.0	22.4
		MCWB	19.2	19.0	19.5	20.8	23.3	24.6	25.8	26.1	24.8	23.4	20.4	20.4
	10%	DB	20.0	20.2	22.3	24.9	28.1	31.7	31.9	32.0	30.5	27.5	22.7	21.1
		MCWB	18.2	18.2	19.1	20.6	23.1	24.6	25.6	26.0	24.7	22.1	19.3	19.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	22.3	22.5	24.9	25.9	28.2	28.5	29.1	27.1	26.3	24.0	23.3
		MCDB	22.2	23.6	24.2	25.7	28.7	32.6	32.5	32.5	30.7	29.1	25.0	23.7
	2%	WB	20.6	21.1	21.6	23.7	25.3	27.3	27.7	28.3	26.4	25.4	22.6	22.2
		MCDB	21.4	22.0	23.2	25.1	28.1	31.1	31.6	31.9	29.9	28.5	24.2	22.9
	5%	WB	19.6	19.9	20.9	22.7	24.8	26.7	27.1	27.6	25.9	24.7	21.4	20.8
		MCDB	20.5	20.8	22.2	24.6	27.7	30.0	30.8	30.7	29.3	27.8	23.2	21.6
	10%	WB	18.6	18.8	20.2	21.9	24.1	26.0	26.6	27.0	25.5	23.8	20.2	19.9
		MCDB	19.9	19.8	21.5	23.6	27.2	29.3	30.1	30.1	29.0	26.7	22.1	20.6
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.3	10.0	9.6	9.2	8.5	8.1	7.9	8.5	10.8	11.7	9.8
		MCDBR	8.4	8.9	8.9	9.4	10.0	9.7	8.7	8.5	9.2	9.7	10.8	9.0
		MCWBR	6.6	6.4	5.4	5.2	4.4	3.6	3.4	3.4	3.8	4.8	7.1	7.2
	5% WB	MCDBR	7.5	7.6	7.5	7.1	7.7	7.8	7.9	7.7	7.8	8.2	9.4	7.6
		MCWBR	6.5	6.4	5.3	5.1	4.1	3.7	3.5	3.5	3.8	5.1	7.4	7.0
Clear-Sky Solar Irradiance	taub	0.336	0.352	0.376	0.407	0.441	0.466	0.508	0.498	0.445	0.387	0.359	0.345	
	taud	2.545	2.503	2.431	2.345	2.286	2.268	2.173	2.197	2.310	2.459	2.475	2.527	
	Ebn at Noon	899	913	910	888	855	829	793	798	830	863	862	868	
	Edh at Noon	86	98	112	126	135	136	150	145	125	101	91	84	
All-Sky Solar Radiation	RadAvg	3.08	3.68	4.97	6.01	6.70	6.32	5.93	5.53	5.20	4.57	3.50	2.71	
	RadStd	0.24	0.50	0.40	0.49	0.34	0.48	0.41	0.41	0.31	0.44	0.31	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+104

Nomenclature: See separate page